

Digital Signal Processing Laboratory

Discrete-Time Fourier Transform (DTFT), FFT, and Convolution Properties

Pouria Amiri
Linkedin GitHub

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Abstract

This report investigates the fundamental frequency-domain representations of discrete-time signals. Through MATLAB simulations, we analyze the properties of the Discrete-Time Fourier Transform (DTFT) including time-shifting, frequency-shifting, and time-reversal. Furthermore, the practical implications of Fast Fourier Transform (FFT) lengths and the execution of linear convolution via circular convolution are computationally verified.

1 Introduction

Transforming signals from the time domain to the frequency domain is a cornerstone of Digital Signal Processing (DSP). The DTFT provides a continuous frequency spectrum for discrete signals, while the FFT offers a computationally efficient discrete frequency representation. This laboratory exercise explores the mathematical properties of these transforms and their applications in system analysis.

2 Properties of the DTFT

2.1 Time and Frequency Shifting

According to Fourier theory, shifting a signal in the time domain does not alter its magnitude spectrum; however, it introduces a linear shift in the phase spectrum. Conversely, a frequency shift translates the magnitude spectrum along the frequency axis and alters the phase. Our MATLAB visualizations confirmed that applying a delay to a sequence preserves the exact shape of $|H(e^{j\omega})|$ while modifying $\arg[H(e^{j\omega})]$.

2.2 Time Reversal

Time-reversing a sequence $x[n]$ to $x[-n]$ affects its Fourier transform by conjugating it (assuming real signals). As demonstrated in the simulation, the magnitude spectrum of the time-reversed sequence is identical to the original sequence, whereas the phase spectrum is exactly inverted (mirrored).



3 FFT Length and Zero-Padding

The resolution of the FFT is heavily dependent on the chosen transform length L relative to the signal length N . A composite sinusoidal signal $x[n] = \cos(0.2\pi n) + \sin(0.4\pi n)$ was evaluated under three conditions:

- $L = N$: The FFT perfectly represents the signal information without data loss, and the Inverse FFT (IFFT) accurately reconstructs the original signal.
- $L > N$ (**Zero-Padding**): Appending zeros to the signal artificially increases the density of frequency samples. This enhances the visual resolution of the frequency spectrum without altering the fundamental physical properties of the time-domain signal.
- $L < N$ (**Truncation**): Results in data loss and spectral leakage, rendering the transformation inaccurate.

4 Circular vs. Linear Convolution via FFT

A key operation in DSP is filtering, governed by linear convolution. However, multiplication in the Discrete Fourier Transform (DFT) domain corresponds to *circular* convolution in the time domain.

To perform linear convolution utilizing the computational efficiency of the FFT, the input sequences must be zero-padded to avoid circular aliasing. For two sequences of lengths N_1 and N_2 , both were padded to length $L = N_1 + N_2 - 1$.

```
1 L_conv = length(x1) + length(x2) - 1;  
2 x1_padded = [x1, zeros(1, L_conv - length(x1))];  
3 x2_padded = [x2, zeros(1, L_conv - length(x2))];  
4 Y_fft = fft(x1_padded) .* fft(x2_padded);  
5 y = real(ifft(Y_fft));  
6
```

Listing 1: Linear Convolution via FFT

The output obtained via IFFT exactly matched the result of MATLAB's native direct `conv()` function, proving that zero-padded circular convolution is mathematically equivalent to linear convolution.

5 Conclusion

This laboratory confirmed the theoretical properties of discrete Fourier analysis. We demonstrated that zero-padding enhances FFT resolution and is absolutely necessary when utilizing DFT/FFT algorithms to execute linear convolution. These principles are vital for implementing rapid digital filters in communication systems.